AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 3 · STUDENT LESSON

Bias terms and decision boundaries

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

For w = [2, -1] and x = [3, 5], what are the contributions and dot product?

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Calculate a weighted score with a bias term. Try the recall before opening notes.

Your learning target

I can calculate a weighted score with a **bias term**.

I can locate and test a two-feature **decision boundary**.

I can explain how changing the bias shifts a boundary without **training**.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Dot products and weighted sums. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Calculate a weighted score with a **bias term**

Locate and test a two-feature **decision boundary**

Explain how changing the bias shifts a boundary without **training**

Retrieve what you already know

> For w = [2, -1] and x = [3, 5], what are the contributions and dot product?

My starting explanation

This week's end goal

By the end, I can explain **bias term**s and decision boundaries using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Bias terms and decision boundaries case. Separate observations, given facts and assumptions before choosing an explanation.

Use the fictional parcel schema [width in cm, height in cm]. Our model computes z = 2 times width - height - 3. The weights are [2, -1]; the bias b is -3. Compute the weighted sum first, then add the bias once.

The result is an affine score: a weighted sum plus a constant offset.

Now declare a classification rule: predict class 1 when z is at least 0, and class 0 when z is below 0. Equality belongs to class 1. These are invented categories for arithmetic practice, not a real parcel decision.

The weights and bias are supplied, so this is a forward calculation, not **training**.

Input [width, height]	Score z	Prediction
[4, 2]	8 - 2 - 3 = 3	1
[2, 4]	4 - 4 - 3 = -3	0
[3, 3]	6 - 3 - 3 = 0	1 at equality

A bias is a mathematical parameter. It is not the same use of the word as unfair treatment; numerical offset alone does not establish whether a system is fair.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Bias terms and decision boundaries. Connect each labelled input, deciding step and result.

Set z to zero to locate the boundary: 2 times width - height - 3 = 0, so height = 2 times width - 3. Plot width on the horizontal axis and height on the vertical axis. Boundary points include (2, 1), (3, 3) and (4, 5).

Join them with a straight line and label the axes in cm.

At width 3, height 2 gives score 1 and class 1; height 3 gives score 0 and class 1; height 4 gives score -1 and class 0. Under this axis convention class 1 lies on or below the boundary.

One point on each side checks that orientation.

```
def classify(width, height, bias=-3):
    score = 2 * width - height + bias
    return score, int(score >= 0)

for height in [2, 3, 4]:
    print(classify(3, height))
# (1, 1), (0, 1), (-1, 0)
```

The code assumes valid numerical measurements in the agreed units. Changing >= to > changes the equality case, even though the drawn line remains in the same place.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Bias terms and decision boundaries. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

For the worked model, find boundary points at width 2 and width 4. Test [4, 4] and [4, 6] to identify the class on each side.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Change bias from-3 to-1.

Predicted result

Prediction: A score becomes2 and B score 0; both predict class 1.

Why the result follows

Reason: Increasing bias by2 adds2 to every score without changing the weights.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Bias terms and decision boundaries to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Bias term	An additive parameter applied after a weighted sum

Word	Meaning
Decision boundary	Input locations where the declared classification rule changes class
Affine function	A weighted sum plus a constant offset
Threshold rule	A stated comparison that converts a score into a class

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

Follow the worked trace for Bias terms and decision boundaries, predict a step, then compare with its explanation.

Hold the weights fixed while shifting the boundary

BIAS	SCORE AT [3, 3]	BOUNDARY EQUATION
b is -3	6 - 3 - 3 gives 0	height = 2 width - 3
b is -1	6 - 3 - 1 gives 2	height = 2 width - 1
Increase b by 2	Every score rises by 2	Slope remains 2

The boundary moves parallel to itself because the weights stay fixed. An increased score does not always change a class: only inputs crossing the declared threshold change.

For [3, 4], compare z and prediction with bias -3 and bias -1. Explain which parameter changed.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Bias terms and decision boundaries

Move a toy boundary with a bias while keeping its **input** weights fixed.

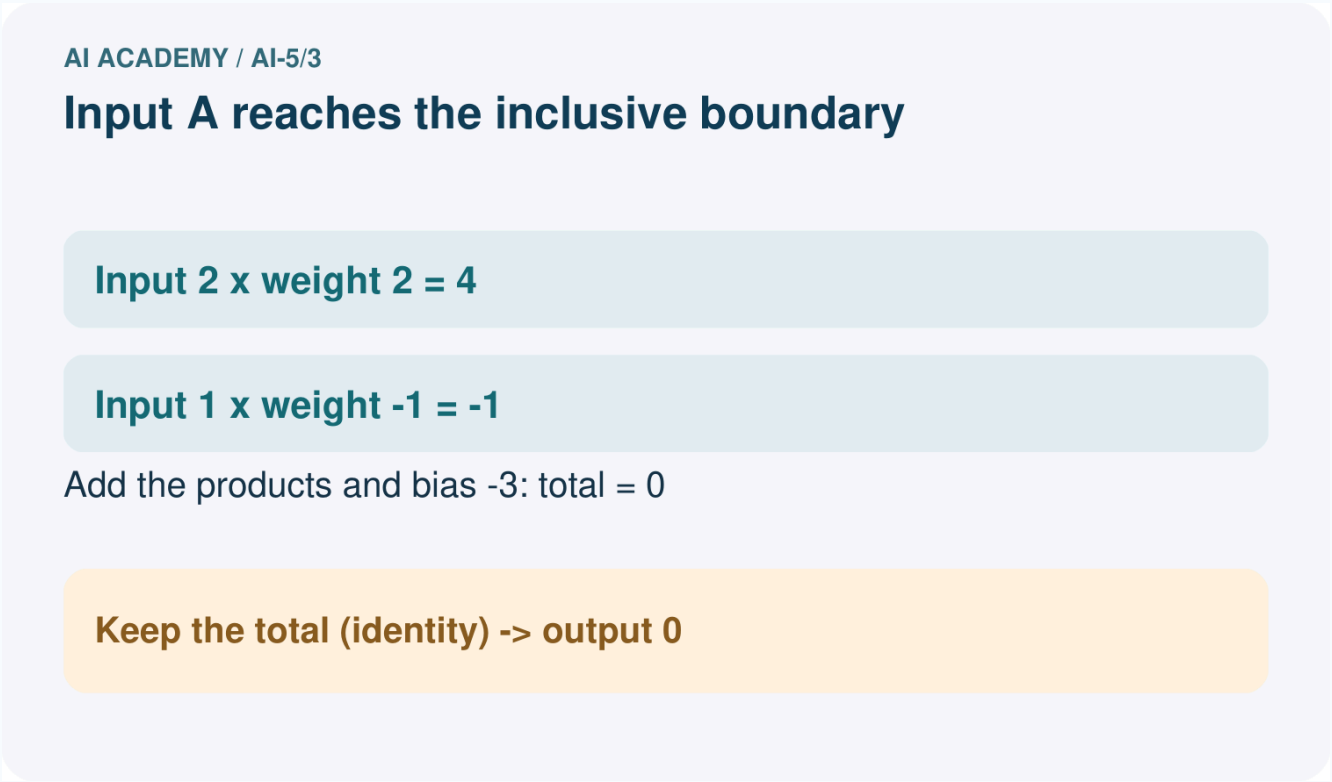
The facts we will use

Supplied facts

Score z=2u-v-3; predict class 1 when z>=0. **Input**sA=[2, 1], B=[1, 1].

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/3

Bias terms and decision boundaries

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Trace input A

A score=4-1-3=0, so class 1 at the declared inclusive boundary.



2. Trace input B

B score=2-1-3=-2, so class 0.



3. Read boundary

The boundary is v=2u-3; the bias sets its position.

CHANGE ONE THING

Change bias from-3 to-1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Trace input A	A score=4-1-3=0, so class 1 at the declared inclusive boundary.
2. Trace input B	B score=2-1-3=-2, so class 0.
3. Read boundary	The boundary is v=2u-3; the bias sets its position.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The bias adds the same offset after the **input** contributions. It moves the dividing line without changing the weights that set the line's orientation. Class 1 includes score zero under our stated rule.

That is why B changes class when the bias rises by two: its score moves from minus two onto the included boundary.

What this example does not establish

Keep the stated limit

A linear boundary is a toy classifier rule. Raw score size does not automatically describe

confidence or probability.

Keep the stated limit

Read the labelled picture: Is B positive under the original or changed model? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Change bias from-3 to-1. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Bias terms and decision boundaries claim and explain a repair using the supplied evidence.

Check the claim against the facts

Without a bias, this score would be z = 2 times width - height and its zero line would pass through (0, 0). The offset lets the boundary move away from the origin.

In this parcel schema zero dimensions are invalid measurements; the origin is still useful as a point on the extended mathematical graph.

Check the claim against the facts

Changing bias from -3 to -1 increases every score by 2, but does not rotate the line. Changing a weight can rotate or otherwise change the boundary. Neither change is automatically an improvement.

To compare models, retain labels, a declared metric, the same evaluation records and the separate **training**, validation and test roles.

Locate the earliest unsupported step

At [3, 3], compare **prediction** under z >= 0 with **prediction** under z > 0. Does changing this comparison move the zero line?

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Bias terms and decision boundaries with support, then explain the deciding step in your own words.

Use z = 2 times width - height - 3 at [5, 4]. Give the weighted sum, bias, final score and **prediction** under z >= 0.

For [3, 4], compare z and prediction with bias -3 and bias -1. Explain which parameter changed.

Explain your prediction to a partner before comparing results.

Practise with less help: Plot and label the boundary

Use workbook W2 for the connected example "**Bias term**s and decision boundaries". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Bias term**s and decision boundaries. Proposed explanation: "Add bias to each contribution". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Is B positive under the original or changed model? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Bias terms and decision boundaries. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

For the worked model, find boundary points at width 2 and width 4. Test [4, 4] and [4, 6] to identify the class on each side.

```
def classify(width, height, bias=-3):
    score = 2 * width - height + bias
    return score, int(score >= 0)

for height in [2, 3, 4]:
    print(classify(3, height))
# (1, 1), (0, 1), (-1, 0)
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Bias terms and decision boundaries. Explain why each step is justified.

Explain → apply → challenge

Use z = 2 times width - height - 3 at [5, 4]. Give the weighted sum, bias, final score and **prediction** under z >= 0.

> At [3, 3], compare **prediction** under z >= 0 with **prediction** under z > 0. Does changing this comparison move the zero line?

A teammate says that raising b must improve the model because all scores rise. What information is needed before accepting an improvement claim?

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled **bias term**s and decision boundaries **evidence** sheet to Image-pattern Network Investigation. Show how you can calculate a weighted score with a **bias term**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

The boundary states what this particular formula predicts. It does not establish the true classes of all possible **input**s.

A point very far from the line need not have a reliable **prediction** when the model, features or data are inappropriate. The raw score is not calibrated confidence or a physical distance from the line.

A team changes b after seeing final-test mistakes and reports the same cases as an untouched test of the new model. Explain the error and the next step.

Next week, we transform scores with activation functions before discussing layers.

A teammate says that raising b must improve the model because all scores rise. What information is needed before accepting an improvement claim?

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Bias terms and decision boundaries case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Hold the weights fixed while shifting the boundary

BIAS	SCORE AT [3, 3]	BOUNDARY EQUATION
b is -3	6 - 3 - 3 gives 0	height = 2 width - 3
b is -1	6 - 3 - 1 gives 2	height = 2 width - 1
Increase b by 2	Every score rises by 2	Slope remains 2

The boundary moves parallel to itself because the weights stay fixed. An increased score does not always change a class: only inputs crossing the declared threshold change.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Analyse a new model independently

A new model has w = [-1, 2], b = -2, **input**s [u, v] and class 1 when z >= 0. Calculate scores and classes for [2, 1], [2, 2], [2, 3]. Write the boundary as v in terms of u. Then change only b to 0 and recompute [2, 1].

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

A new model has w = [-1, 2], b = -2, **input**s [u, v] and class 1 when z >= 0. Calculate scores and classes for [2, 1], [2, 2], [2, 3]. Write the boundary as v in terms of u. Then change only b to 0 and recompute [2, 1].

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Bias terms and decision boundaries, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Move a toy boundary with a bias while keeping its **input** weights fixed.

By the end, I can	How I will show it
Calculate a weighted score with a bias term	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Locate and test a two-feature decision boundary	Explain the deciding step, show a trace or test, and state one limit.
Explain how changing the bias shifts a boundary without training	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Bias terms and decision boundaries without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Calculate a weighted score with a **bias term**

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Move a toy boundary with a bias while keeping its **input** weights fixed.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The boundary is v=2u-3; the bias sets its position.

1. Trace **input** A

2. Read boundary

3. Trace **input** B

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Read boundary The boundary is v=2u-3; the bias sets its position. The bias adds the same offset after the **input** contributions.

It moves the dividing line without changing the weights that set the line’s orientation. Class 1 includes score zero under our stated rule.

That is why B changes class when the bias rises by two: its score moves from minus two onto the included boundary.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change bias from-3 to-1. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. A score becomes2 and B score 0; both predict class 1.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: A score becomes2 and B score 0; both predict class 1. Increasing bias by2 adds2 to every score without changing the weights.

Repair a missing building block

Review Dot products and weighted sums (ai-5/2). Goal: Calculate a dot product by matching feature positions. Short practice: Which connection explains why more right brightness lowers this score?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

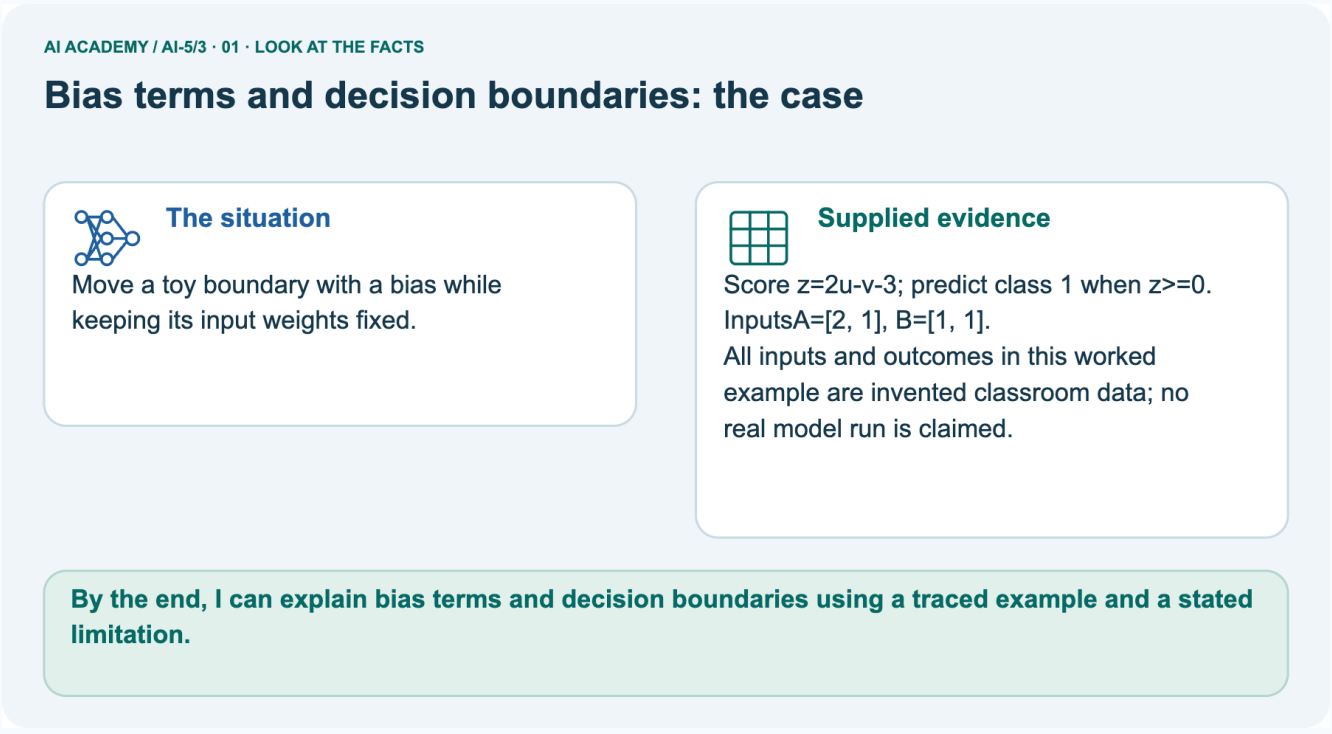
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Bias terms and decision boundaries graphic using its labels, deciding evidence and relevant limitation.

01. Bias terms and decision boundaries: the case



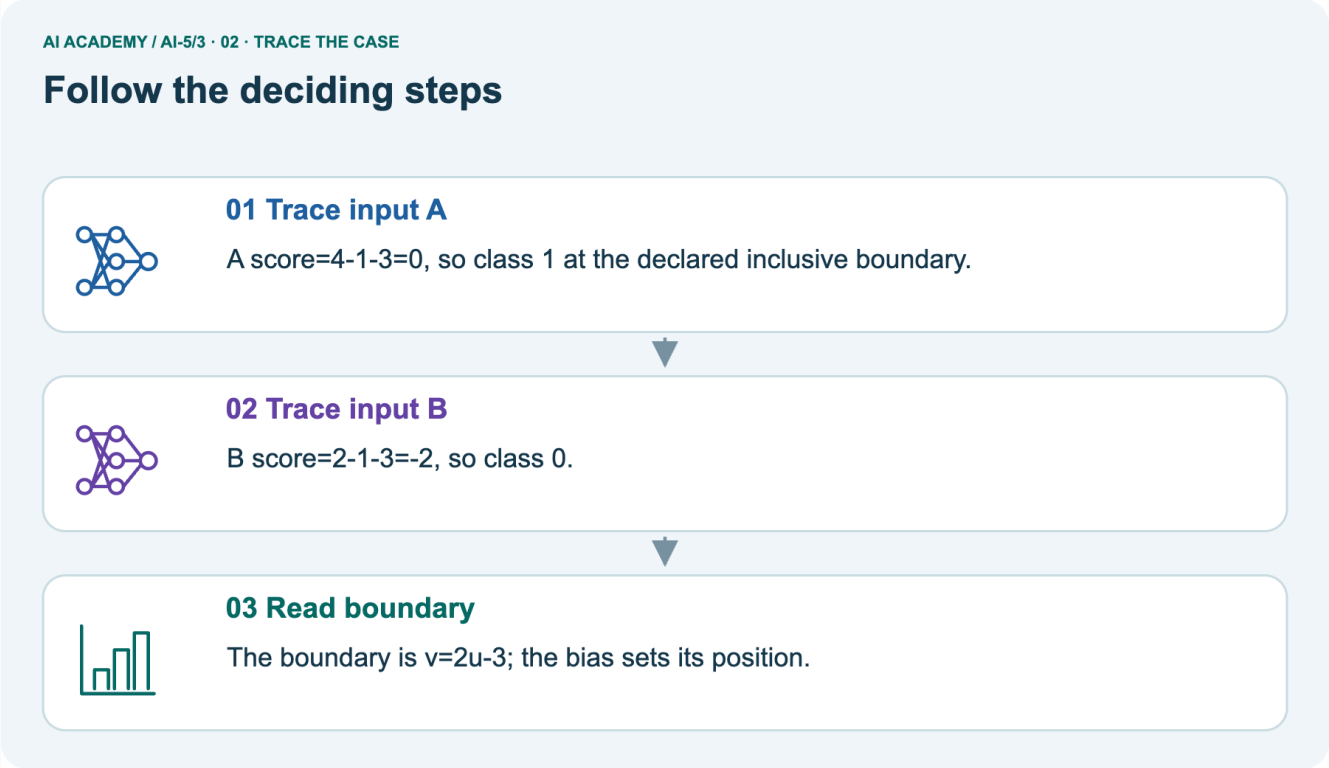
Follow the labelled picture with the matching explanation. Open the vector diagram

Move a toy boundary with a bias while keeping its **input** weights fixed. Supplied facts: Score z=2u-v-3; predict class 1 when z>=0. **Input**sA=[2, 1], B=[1, 1].

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

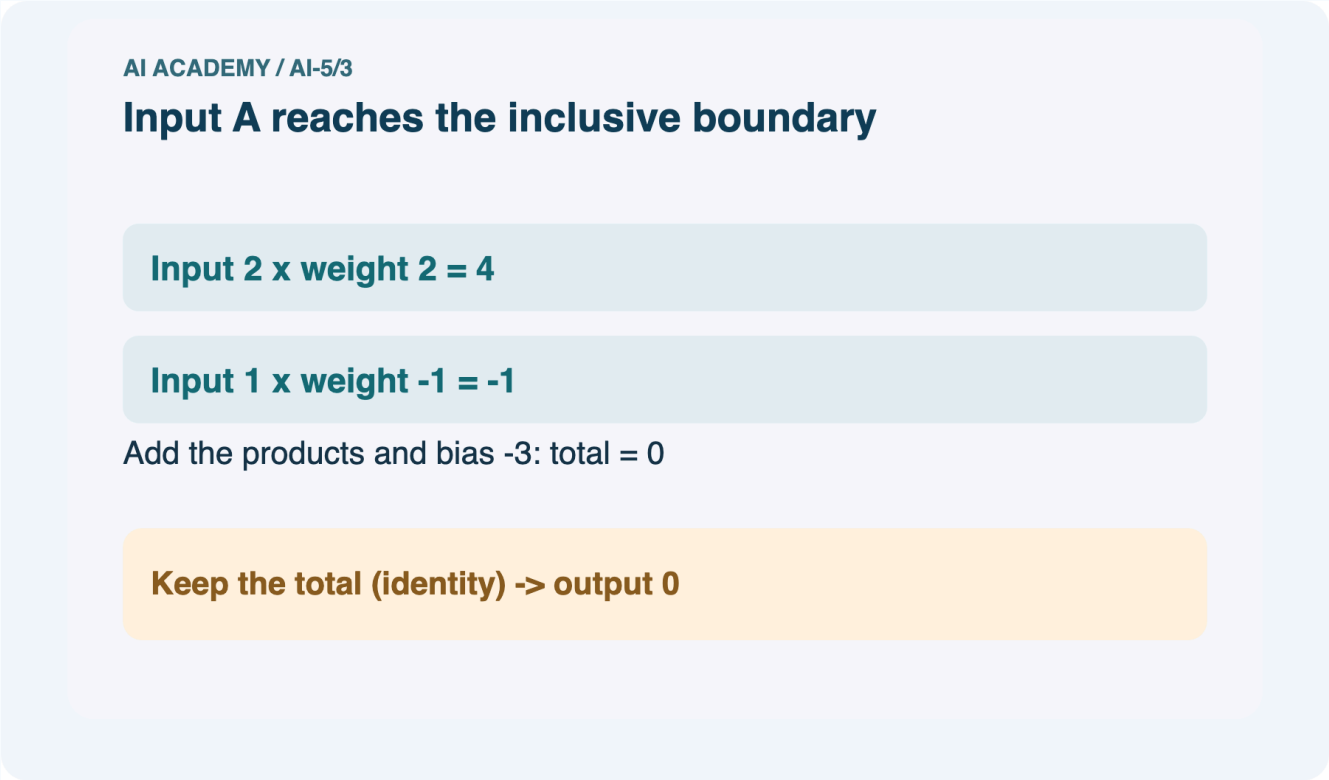


Follow the labelled picture with the matching explanation. Open the vector diagram

Trace **input** A: A score=4-1-3=0, so class 1 at the declared inclusive boundary. → Trace **input** B: B score=2-1-3=-2, so class 0. → Read boundary: The boundary is v=2u-3; the bias sets its position.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Input A reaches the inclusive boundary

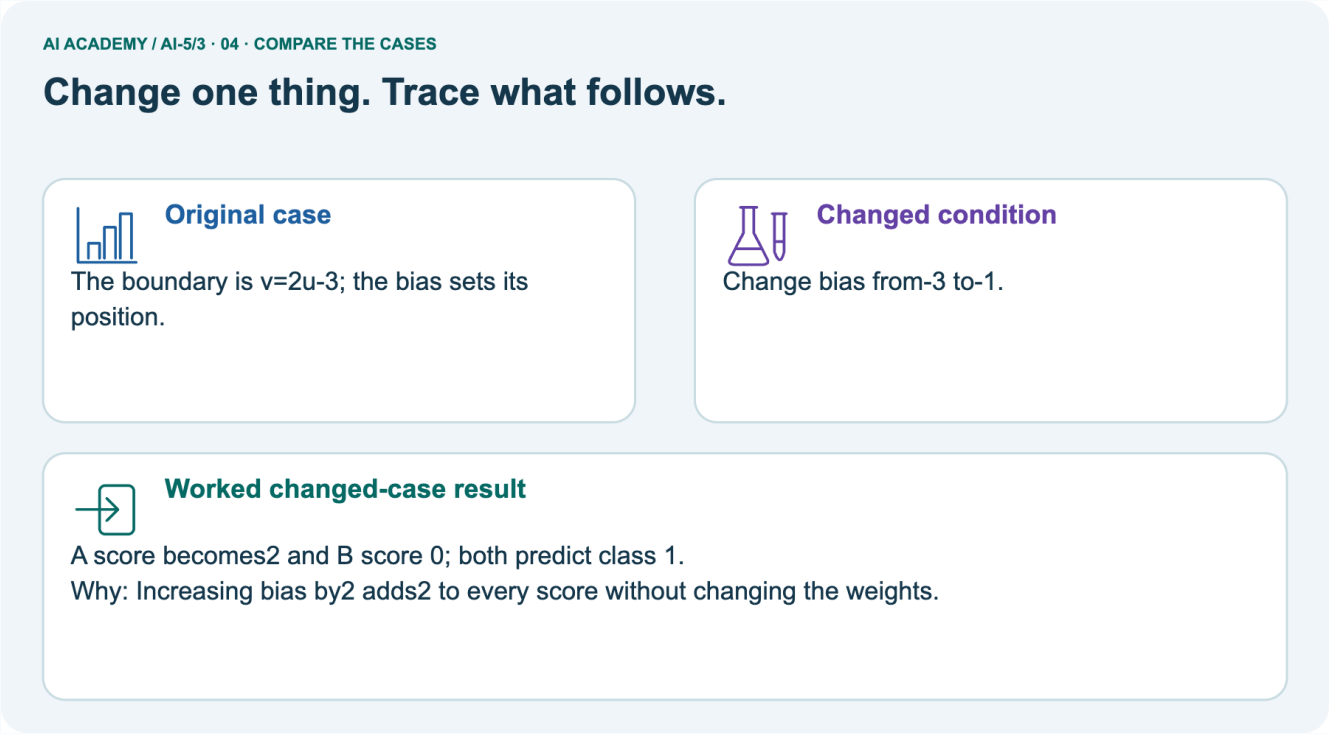


Follow the labelled picture with the matching explanation. Open the vector diagram

Input A reaches the inclusive boundary. **Input**s: [2, 1]; weights: [2, -1]; bias: -3; activation: identity; **output**: 0.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



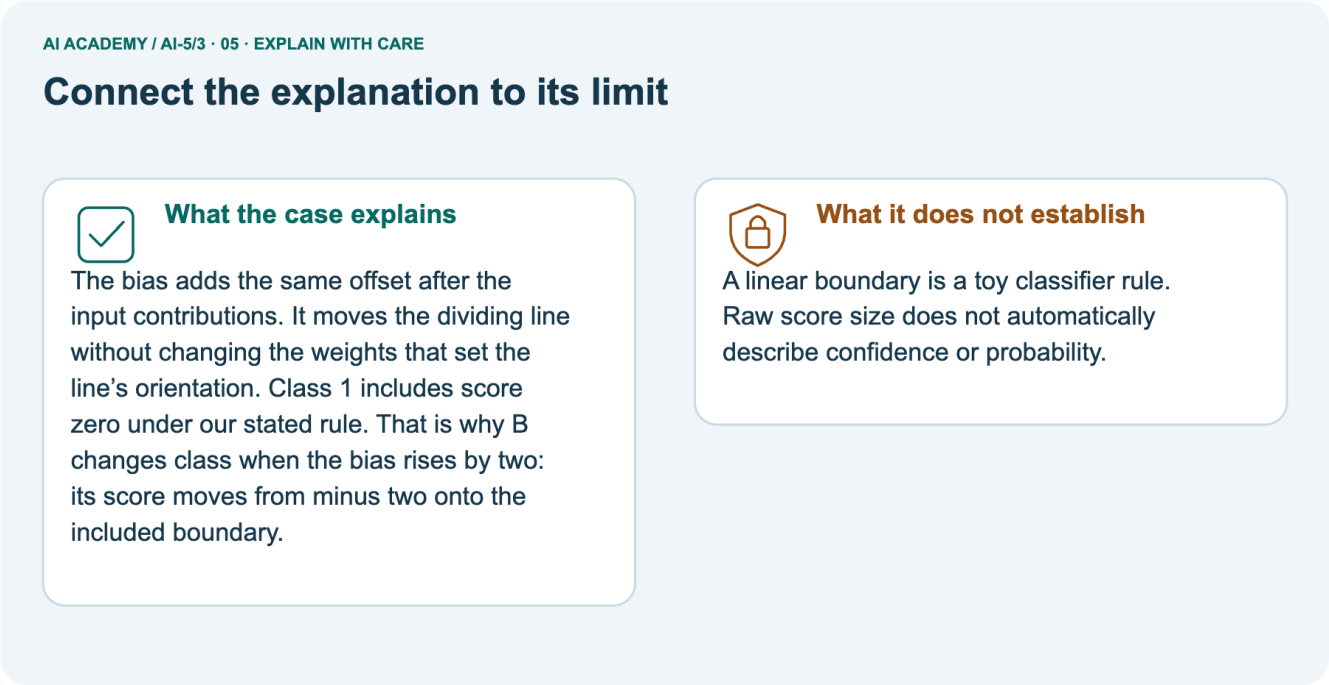
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The boundary is v=2u-3; the bias sets its position. Changed condition: Change bias from-3 to-1. Expected result: A score becomes2 and B score 0; both predict class 1.

Reason: Increasing bias by2 adds2 to every score without changing the weights.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

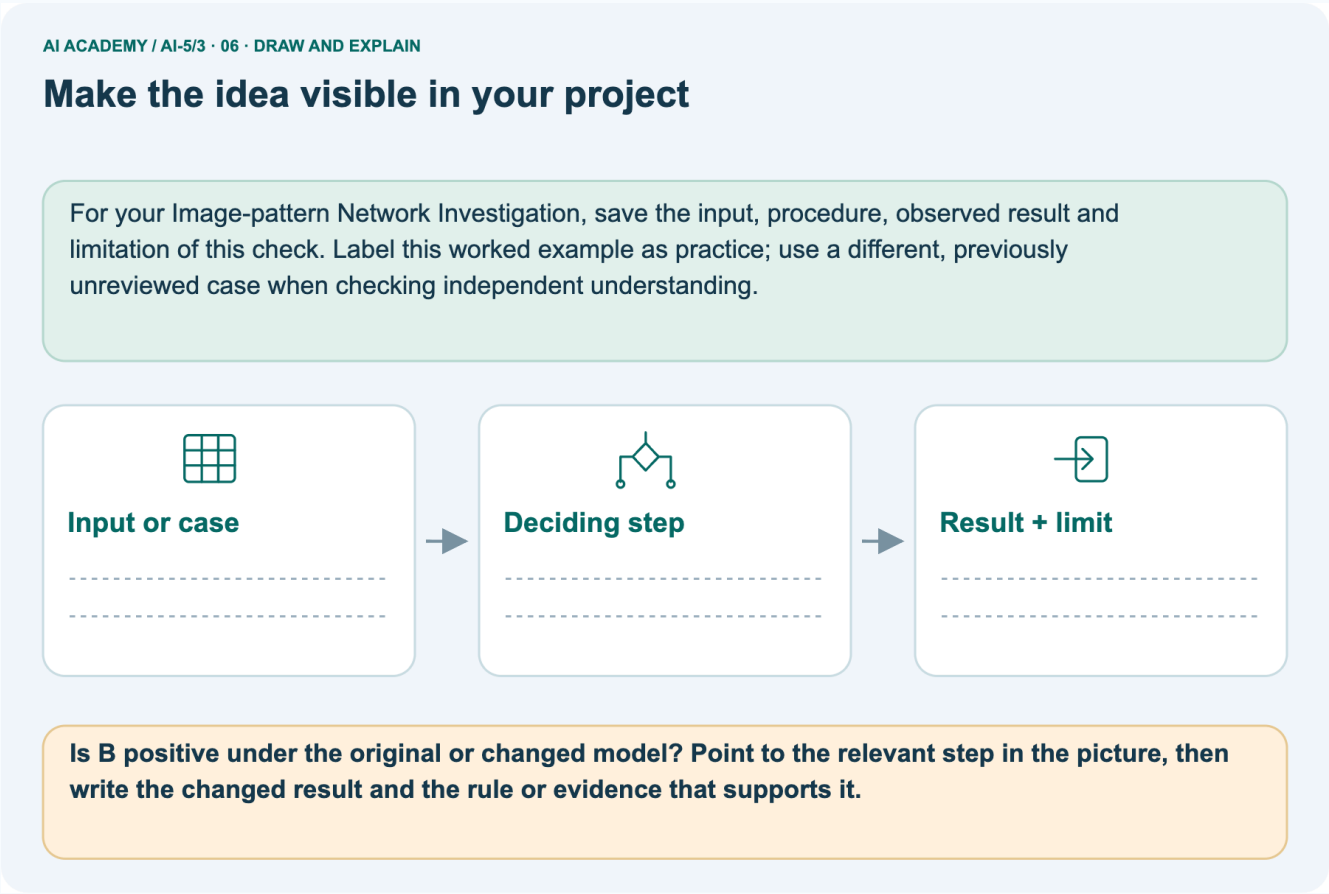
The bias adds the same offset after the **input** contributions. It moves the dividing line without changing the weights that set the line's orientation. Class 1 includes score zero under our stated rule.

That is why B changes class when the bias rises by two: its score moves from minus two onto the included boundary. Limit: A linear boundary is a toy classifier rule.

Raw score size does not automatically describe confidence or probability.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Is B positive under the original or changed model? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.